Polar Bayesian decoding: complete decoder options

From exact identities and approximation algebra to full decisions

19 October 2026 — Report 10; design and derivation, before implementation

Purpose. Assemble the receiver options implied by Reports 7 and 9. A useful decoder description must reach full-message counts from the observation, including discovery, normalization, multiplicity and a stopping rule. A moment formula or a local update alone does not do that. We retain one exact reference family and thirteen execution families, with shared variations. This covers the existing mathematical resources, not every possible future algorithm or combination.

Status. These are mathematical designs, with explicit finite constructions and conditional guarantees. None has been implemented or ranked in this report. Several remain exponentially expensive. The new derivations close product-law discovery, auxiliary-field importance correction and a sequential particle option. Report 7 continues to own exact foundations; Report 9 owns approximation bounds A–T; Report 8 supplies the frozen encoder. The design log is `docs/POLAR_DECODER_DESIGN_LOG.md`. The identifiers DEC-01–DEC-13 are unrelated to the historical D0–D4 receiver ladder.

1 The same problem and decision for every option

The known encoding rule maps a full message $w = (p, v)$ to

$$\phi(p, v) = S_p \operatorname{diag}(\sqrt{e_p})(-1)^{Lv}, \qquad \|\phi(w)\| = 1, \qquad y = \sum_{k=1}^{K} \phi(w_k) + z, \quad z \sim \mathcal{N}(0, \sigma^2 I_n).$$

Patterns and payloads are unknown. $P = 2^{B_p}$, $V = 2^{B_c}$ and $M = PV = 2^B$. The receiver knows K, $\sigma^2 > 0$, and the fixed polar/CRC, placement and balanced-energy construction. No option changes the physical noise or assumes the active patterns are supplied.

Under the primary IID uniform source, a count vector a and an ordered tuple W have weights

$$\omega(a) = \frac{K!}{M^K \prod_w a_w!}\, g(y - \Phi a), \qquad \omega_{\text{slot}}(W) = M^{-K} g\left(y - \sum_k \phi(w_k)\right), \qquad g(r) = e^{-\|r\|^2/(2\sigma^2)}. \quad (1)$$

Both normalize by the same evidence Z. Summing ordered tuples with the same counts gives the factorial weight in the first expression. A sampling proposal need not be exchangeable; the target still is. Do not insert a second factorial factor into ordered-slot weights.

For an output d with nonnegative integer counts and $\sum_w d_w = K$, the loss is

$$\ell(a, d) = 1 - \frac{1}{K} \sum_w \min(a_w, d_w).$$

The sufficient decision statistics are count tails $T_{w,j} = \Pr(a_w \geq j \mid y)$, $1 \leq j \leq K$. Choose the largest available next tail T_{w,d_w+1} until K copies have been allocated. This greedy rule is exact because each word's tails decrease. MAP, a single sampled tuple and top-K posterior means are generally different decisions.

The distinct-message control uses ordered prior $1/(M)_K$ on injective tuples, or uniform prior $1/\binom{M}{K}$ on 0/1 counts. Require $K \leq M$ and output distinct words there. It retains collisions of patterns with different payloads. Every extension to it below keeps those exclusions explicitly; an IID approximation theorem is not automatically a distinct-source theorem.

1.1 What makes an option a full decoder

Each option specifies: a finite construction from the known encoder and y; how full words are discovered or enumerated; the legal law or exact queries being evaluated; count-tail readout; a feasible allocation and stopping rule; and the cost/error that remains. Computational infeasibility is permitted at this design stage, but uncharged inner or posterior oracles are not.

For later implementations the common interface remains

```
run(scenario, **params) -> (counts, meta).
```

Here `counts` may be a sparse full-word map internally; a mandatory dense M-vector adapter would itself cost M and must be counted. The mathematical output contract does not require that adapter. Relevant metadata are the source, variant, resource budget, realized construction/query work, target-bias bound, numerical error/failure budget and conditional excess-loss certificate. No implementation is added now.

2 A map of the options

The numbering is an index, not a performance ordering. Start with inexpensive inner formulas when a family permits them, then refine only as its accuracy requires.

ID	Core idea	How it reaches actual decoded words
REF	Exact posterior evaluation	Sum legal explanations, obtain count tails and allocate copies. Different exact representations compute the same decision.
DEC-01	Direct residual queries	Visit every word and estimate its count masses from excluded-word residual likelihoods.
DEC-02	Variational product law	Optimize one common one-user law; discover its highest-probability words and use binomial count tails.
DEC-03	Message-prefix refinement	Bound the recoverable gain inside unresolved sets of words; split only cells that can affect the allocation.
DEC-04	Configuration search / bounded list	Search explanations of all K users; retain complete tuples and bound the mass of the unresolved frontier.
DEC-05	Independent guided importance	Draw full tuples using approximate continuation scores; exact weights correct their guidance, and weighted counts give tails.
DEC-06	Corrected local/global chain	Combine exact-target global proposals with local one-, two- or three-user moves; use count frequencies after a justified schedule.
DEC-07	Sequential particles	Grow and resample clouds of partial tuples; terminal physical-likelihood correction and independent batches yield count queries.

ID	Core idea	How it reaches actual decoded words
DEC-08	Quantized continuation states	Merge compatible partial sums with positive corrections, then reverse the recorded graph to recover count marginals.
DEC-09	Positive factor contraction	Retain regional counts and exact constraints; simplify interactions or condition feature indices, then contract count-marked factors.
DEC-10	Connected-user expansion	Sum small connected blocks, including all their combinations across K slots, and reconstruct marked count polynomials.
DEC-11	Fourier query decoder	Replace residual convolutions by frequency calculations; bound signed count numerators and combine with full-word traversal or prefix search.
DEC-12	Signed low-rank contraction	Compress likelihood tables and contract their positive error envelopes alongside count marks; use bounded queries for the final decision.
DEC-13	Positive auxiliary fields	Integrate shared Gaussian fields and independent conditional one-user laws, or sample the extended field/tuple space with exact correction.

These families overlap deliberately at interfaces. DEC-03 is a discovery/decision strategy, and can use DEC-01, 10, 11, 12 or 13 as query providers. DEC-05 can use a law from DEC-02, 08 or 09 as its proposal when its density and draws are available. Changing an inner moment order is a variation; changing from message-prefix queries to configuration sampling changes the execution family. The combination rules near the end specify which connections are legitimate.

3 Shared readout, stopping and error rules

3.1 From tails or intervals to a feasible action

If estimated tails have uniform error e and the action maximizes their gain, Report 9 gives

$$R(\widehat{d} \mid y) - R(d^* \mid y) \le \min(1, 2e). \tag{2}$$

Clip estimates to $[0,1]$ and sort each word's tails in decreasing order if necessary. Order statistics cannot increase uniform distance from the true sorted tails, so this preserves the bound and makes greedy allocation valid. It does not assert joint consistency of unrelated word estimates.

For interval-only calculations, let $[l_{w,j}, u_{w,j}]$ contain every exact tail. Choose a feasible incumbent maximizing lower-tail gain, using monotone closures of valid bounds. For each word these closures are $l_j \leftarrow \max_{r \ge j} l_r$ and $u_j \leftarrow \min_{r \le j} u_r$, after intersection with $[0,1]$. An upper bound on any feasible gain is obtained by maximizing the upper-tail gain under the same budget. If these bounds are L_* and U_*, then

$$R(\widehat{d} \mid y) - R(d^* \mid y) \le \min\{1, (U_* - L_*)/K\}.$$

DEC-03 extends this rule to unresolved prefixes. At a resource limit, return the incumbent with the remaining bound, possibly one. A prefix, latent state or field sample is never itself a recovered word. Ties may prevent strict optimality certification; an arbitrary positive excess-risk tolerance is a different stopping requirement.

3.2 A common independent-replicate guarantee

Several designs produce independent replicates $(Y_s, X_{s,H})$ for bounded count observables H, with

$$0 \le X_{s,H} \le Y_s \le C, \qquad \mathbb{E}Y_s = Z_0 > 0, \quad \mathbb{E}X_{s,H} = Z_0 T_H.$$

Z_0 may be the physical evidence, a fixed rescaling, or an explicitly restricted target evidence. For J queries and failure budget δ, put

$$b_S = C\sqrt{\frac{\log[2(J+1)/\delta]}{2S}}, \qquad \widehat{T}_H = \frac{\overline{X}_H}{\overline{Y}}.$$

The bounded-variable exponential inequality and a union bound give simultaneous numerator/denominator errors at most b_S. Hence, when $\overline{Y} > b_S$,

$$\max_H |\widehat{T}_H - T_H| \le e_S := \min\left\{1, \frac{2b_S}{\overline{Y} - b_S}\right\}; \quad \text{otherwise use } e_S = 1. \tag{3}$$

If $\overline{Y} = 0$, do not divide: retain a fixed legal incumbent and the trivial risk bound one, then continue or stop at the declared budget. For IID sampling the incumbent can be K copies of one fixed word; for the distinct control it is the first K legal words. Take $J = MK$ for all word tails, including unseen words, or $J = K(2M-1)$ for all binary message-prefix count tails. The logarithm is computable from B, K without constructing those arrays. Weights may all be rescaled together; C must use the same scale. It is a proved global bound, not the largest observed weight. Report 9 H also gives variance-sensitive sample rates.

A positive weighted sample law gives monotone tails automatically. A sample containing count c of a word updates its first c tails, so only K updates are needed per complete tuple. Unseen words have zero empirical tails but remain covered by the simultaneous error bound. For prescribed rounds, allocate $\delta_r = \delta/[r(r+1)]$. This permits stopping on a certificate. Freeze a learned-from-computation proposal before using fresh independent draws, or supply a separate adaptive proof; numerical independence cannot be inferred from an effective-sample-size diagnostic.

3.3 Keep target approximation separate from numerical uncertainty

If support restriction costs posterior mass η, a positive conditional target approximation has TV error τ, and its normalized tails have numerical error e, then a valid total error is

$$e_{\text{total}} \le \eta + (1-\eta)\min(1, \tau + e). \tag{4}$$

For compatible positive weight modifications with log-ratio range Ω, one may use $\tau = \tanh(\Omega/4)$. Exact importance correction removes proposal bias, while retaining sampling error and concentration cost. Signed computations instead supply numerator/evidence intervals. Do not count a corrected guide error again as posterior bias, or omit an error in the target weights. All bounds here are in real arithmetic; certified finite-precision enclosures are a later implementation duty.

4 REF: exact finite Bayes decoding

In plain terms. Give every legal explanation its physical likelihood and source probability, then choose the output with greatest expected number of correctly recovered copies. This is the reference against which an approximation's query and decision errors can eventually be measured.

A direct construction visits the $H(M,K) = \binom{M+K-1}{K}$ IID count vectors, accumulates Z and each occupied word's marked count masses, normalizes and allocates copies. It requires at most K positive count-tail updates per configuration. Computing each signal directly costs $O(KC_{\text{enc}} + nK)$; enumerating $H(M,K)$ configurations remains the dominant issue. Streaming avoids storing every posterior state, but an all-word tail table can require $O(MK)$ storage.

Report 7 supplies equivalent execution variants:

1. Excluded-alphabet residual recursion, with a Gaussian base case.
2. Positive inside/outside convolutions over message-prefix alphabets, retaining multiplicities.
3. Exact projected/frontier schedules and resource finalization.
4. Polar/pattern factor elimination with joint separators and count counters.
5. Full Fourier inversion or the full connected-user expansion.
6. Specialized exact polar pair and triangle contractions for $K = 2, 3$.
7. Certified partial summation that stops before exhaustive evaluation if the Bayes decision separates.

These are not different optimal decoders statistically. Their cost, reuse and intermediate objects differ. The low-dimensional parity/projection structure can reduce operations without removing explicit alphabet, joint-interface or reachable-state growth. Equal-signal classes retain source-weighted word identities: replacing a class by an arbitrary representative would change the loss.

5 DEC-01: direct residual-count decoding

In plain terms. For each possible word, ask how plausible the observation is if that word appears zero, one, two, or more times. Explain the remaining signal using the other words.

For $M > 1$, let $X^{(-w)}$ be uniform on all words except w, and define

$$L_h^{(-w)}(r) = \mathbb{E}g\left(r - \sum_{i=1}^{h} X_i^{(-w)}\right).$$

The exact count numerators are

$$N_{w,j} = \binom{K}{j} M^{-j}(1 - 1/M)^{K-j} L_{K-j}^{(-w)}(y - j\phi_w), \qquad \sum_{j=0}^{K} N_{w,j} = Z. \tag{5}$$

Exclusion is essential: leaving w in the remainder produces factorial moments instead. $M = 1$ is deterministic. A finite decoder visits every word, evaluates or bounds these $K + 1$ masses, normalizes per word, takes tails and applies Section 3.

Retained variations. Use F's Gaussian smoothed-residual formula; a mixture over explicit modes; N's tilted closure with Q's finite inner queries; or Q's positive likelihood-polynomial hierarchy. The latter converges with increasing degree without requiring Gaussian interference. Alternatively query M's factorial moments, with an *unrestricted* remainder, and reconstruct tail intervals using P's finite moment LP. At full order exact moments determine each word's count law. These changes affect the query provider, not the decoder's full-alphabet traversal.

Gaussian masses are positive and yield monotone per-word tails, though their separate normalizers need not be one common approximate evidence. Use direct tail bounds, not an invented joint posterior TV. For signed providers, bound event and complement masses before division. Shared-evidence and moment constraints can tighten the intervals without assuming independent errors.

Cost and stopping. If C_L includes a residual query's inner moments, partitions and matrices, direct work is $O(MKC_L + MK)$ plus tail selection and encoder construction, with possible shared work. Excluded moments use the exact rank-one correction in F. Complete readout is already explicit in M. Choose a query tolerance/degree or return its actual bound at a budget limit. A Gaussian closure alone need not approach exact inference as numerical precision increases; use the polynomial or exact residual variant when a convergent refinement is required. Under distinct sampling, only counts zero and one remain, with the separate excluded distinct residual and masses from Report 9's completion section.

6 DEC-02: an optimized product-law decoder

In plain terms. Approximate the joint posterior by K independent draws from one optimized message distribution. This is the simplest retained joint law, but it can miss strong dependence.

For $\psi(\theta) = M^{-1}\sum_w e^{\theta^T\phi_w}$ and $\mu = \nabla \log \psi$, S specifies

$$\nu_\theta(w) = \frac{e^{\theta^T\phi_w}}{M\psi(\theta)}, \qquad Q_\theta(W) = \prod_k \nu_\theta(w_k), \qquad \sigma^2\theta_* + (K-1)\mu(\theta_*) = y.$$

Optimize the strictly convex tilt objective with S's finite gradient accuracy bound, or stop at a declared trial tilt. Q's character polynomials or explicit sums evaluate its partitions/moments. The true objective value and an evidence upper bound give

$$\tau_{\text{var}} \le \min\{1, \sqrt{(\log Z_+ - F_-)/2}\}.$$

This certifies the chosen reference only when informative. Exact optimization is not a guarantee of a good approximation; Report 9 gives a two-user counterexample.

6.1 Close discovery using the highest-probability words

Under Q_θ, $a_w \sim \text{Binomial}(K, \nu_\theta(w))$. Every binomial tail is increasing in its probability, and $T_{w,j} \le T_{w,1}$. Let C contain the $\min(K, M)$ highest-probability words. If an allocation uses a word outside C, then some word in C has no allocated copy: there are only K copies in total. Moving an outside copy to that inside word's first copy does not decrease gain, since its first tail is at least the removed outside tail. Repeat to obtain an optimal allocation supported on C. Ties can be resolved deterministically. This is a property of this product law, not a top-K shortcut for the physical interacting posterior.

The probabilities are ordered by $s(w) = \theta^T\phi(w)$. For a fixed pattern and conditioned payload prefix, Q's character grouping gives $s(z) = a_0 + \sum_{g\neq 0} a_g\chi_g(z)$ and upper bound $a_0 + \sum_g |a_g|$. Initialize the P pattern roots, split the highest-bound payload prefix, and evaluate full words at leaves. Stop when no unresolved bound exceeds the $\min(K, M)$th retained score; include tie-breaking in that comparison. Full traversal is the finite worst case. Pattern enumeration and key construction are charged.

Evaluate the normalizer and binomial tails on C, then allocate K copies by next-tail gain. With exact partitions this is the exact Bayes action for Q_θ. Certified partition intervals instead bound each $\nu_\theta(w)$; binomial-tail monotonicity propagates them. Probability error ϵ_p gives tail error at most $K\epsilon_p$ by coupling K Bernoulli draws. Add numerical error to τ_{var} or use direct allocation intervals.

Variations and cost. The one-user tilt $\theta = y/\sigma^2$ is an unoptimized variant. One common positive Taylor weight across words gives a coherent polynomial product surrogate; its additional source-law error must be charged. Costs include all optimization queries, P roots, visited prefixes, the partition and at most K word-count distributions. The search or partition can still enumerate all M words. This product law is IID-specific; conditioning it on distinctness requires a separate coupled calculation.

6.2 A complete slot-specific product variation

The larger family $Q(W) = \prod_k \nu_k(w_k)$ is also finite, but its optimization is generally nonconcave. Write $\mu_k = \mathbb{E}_{\nu_k}\phi$, and let u be the uniform one-word law. Its evidence objective is

$$F(Q) = -\frac{\|y - \sum_k \mu_k\|^2 + \sum_k (1 - \|\mu_k\|^2)}{2\sigma^2} - \sum_k D_{\mathrm{KL}}(\nu_k \| u).$$

Holding the other laws fixed, the exact coordinate optimum is

$$\nu_k(w) \propto e^{\theta_k^T \phi_w}, \qquad \theta_k = \frac{y - \sum_{i \neq k} \mu_i}{\sigma^2}.$$

Start, for example, from zero tilts and perform a declared number of cyclic coordinate sweeps. Exact coordinate updates cannot decrease F, but give no global optimum guarantee. With approximate inner moments, treat the resulting tilts as trial references and bound their actual objective; an optimization improvement is not assumed.

For each word, obtain its count probabilities from $\prod_k [1 - \nu_k(w) + z\nu_k(w)]$ in $O(K^2)$ arithmetic. Full alphabet traversal, followed by tails and greedy allocation, closes readout; the identical-slot top-K discovery proof is not reused. Alternatively these product cell-count queries can feed DEC-03, with the variational error allowance included. The same evidence-gap TV certificate applies by marginalization, although slot/count KL need not be equal when the factors differ. The variation costs the coordinate partition/moment work plus $O(MK^2)$ direct readout. It can represent different likely roles for slots, but remains an independent product and can still miss essential posterior correlations. Symmetrizing slot labels changes no count law or count decision.

7 DEC-03: decision-directed message-prefix refinement

In plain terms. Keep large sets of possible words unresolved until their possible contribution to the output makes them worth splitting. Ask count questions about each set rather than guessing one word.

Maintain a disjoint binary prefix cover $\mathcal{C}$ of the *message* alphabet. For a cell D, let $A_D = \sum_{w \in D} a_w$ and bound

$$G_D(h) = \mathbb{E} \min(A_D, h), \qquad h = 0, \ldots, K.$$

For any allocation of h output copies inside D, its matching gain is at most $\min(A_D, h)$ pointwise. If $U_D(h)$ bounds $G_D(h)$, then

$$U_* = \max_{\substack{h_D \geq 0 \\ \sum_D h_D = K}} \sum_{D \in \mathcal{C}} U_D(h_D) \tag{6}$$

is an upper bound on the best physical-posterior gain. A budget dynamic program costs $O(|\mathcal{C}|K^2)$; concavity may simplify special cases but is not assumed. Bounds $0 \leq G_D(h) \leq h$ always initialize the calculation.

Resolve one legal word initially and assign its K copies for an IID incumbent; for distinct sampling resolve K distinct words. Retain lower tail bounds on resolved leaves to get incumbent gain L_*. Refine a cell used by an optimistic allocation in (6), or tighten a query whose uncertainty controls the gap. Stop when $(U_* - L_*)/K$ is small enough, or return the actual remaining bound. Resolve full words before output; an unresolved cell is only an upper bound.

Query variants. Use conditional polar moments and excluded-cell residuals; tilted or polynomial residual queries; marked connected, Fourier or signed-contraction queries; field count polynomials; or simultaneously bounded configuration samples. P's moment LP, shared evidence and prefix split tables combine available information. They cannot remove the proven ambiguity of insufficient moments. Gaussian-only queries can stall. A declared convergent provider or exact subtree summation closes the finite refinement route. A deterministic fair schedule can interleave promising refinements with the oldest unresolved cell/query; budget termination does not require full refinement.

Cost. Charge every visited prefix and every provider invocation, its preprocessing, the current cover program and full-word readout. There can be $2M - 1$ message-prefix nodes. If a randomized provider is used adaptively, cover all possible prefix queries or allocate conditional failure budgets to fresh queries. Choosing favourable estimates after seeing them does not preserve a fixed-query guarantee.

8 DEC-04: bounded configuration search and list decoding

In plain terms. Build complete explanations of all users. Keep exact weights for explanations already completed and a rigorous upper bound on everything still unexplored. This differs from DEC-03: its prefixes belong to the KB-bit *ordered tuple*, not one message.

Partition the tuple space into evaluated leaves $\mathcal{L}$ and a disjoint unresolved prefix frontier $\mathcal{F}$. A node ξ has prior mass $\pi(\xi)$ and conditional likelihood $L(\xi) = \mathbb{E}_\pi[g(y - s_W) \mid \xi]$. Any valid $u_\xi \geq L(\xi)$ gives

$$Z_\mathcal{L} = \sum_{W \in \mathcal{L}} \omega_{\text{slot}}(W), \qquad U_\mathcal{F} = \sum_{\xi \in \mathcal{F}} \pi(\xi) u_\xi, \qquad \eta \leq \frac{U_\mathcal{F}}{Z_\mathcal{L} + U_\mathcal{F}}.$$

The bound follows from the unknown omitted evidence being at most $U_\mathcal{F}$. Use $u_\xi = 1$ at worst, or Report 7's convex-completion/support bounds and Report 9's likelihood intervals. Those are bounds on unexplored completions, not substitutes for their total prior mass.

Initialize by completing one legal tuple and retaining the sibling prefixes along its path. This supplies positive $Z_\mathcal{L}$ and a cover of every other tuple. Expand a frontier node, replace it by its legal children and update the bounds. Complete leaves use the physical likelihood exactly. Their normalized positive weighted counts give a feasible Bayes action on the retained list, with excess risk at most 2η, capped at one. Marked adverse-event bounds can give a sharper decision certificate. Multiple orderings of one count vector must all retain their mass.

Variations. Best-first mass-bound expansion, depth-first traversal and a bounded beam/list change exploration order or memory. A discarded beam node still belongs to the omitted frontier: keep its bound, or a valid aggregate bound, instead of silently declaring it impossible. A message-list restriction is another event and needs a bound for configurations containing omitted words. Report 7's count-tree version avoids ordered duplicates using its exact factorial prior.

Cost and limit. In the worst case the ordered search has M^K leaves. Each bound/leaf evaluation and frontier record is charged. The K sparse updates per completed tuple avoid an additional full alphabet readout. Exhaustion gives REF; a limited list gives an approximate decision with its actual unresolved bound. Keeping only the best physical residual and returning it would instead be a MAP-like decision and is not this decoder.

9 DEC-05: independent guided importance decoding

In plain terms. Let a cheap approximation suggest complete tuples, then correct each suggestion using its exact physical likelihood and the probability with which it was proposed.

At an ordered-tuple bit prefix ξ, choose a positive continuation guide $\widetilde{L}$ and use

$$q_g(b \mid \xi) = \frac{\pi(b \mid \xi)\widetilde{L}(\xi b)}{\sum_c \pi(c \mid \xi)\widetilde{L}(\xi c)}, \qquad q_g(W) = \prod_{t=1}^{KB} q_g(b_t \mid \xi_{t-1}).$$

The prior branch is $1/2$ for IID uniform bits. Two child queries per bit suffice; both pattern and payload bits are generated. F's conditional Gaussian guide is one starting choice; N/Q, positive contractions or exact small-block sums give alternatives. If certified child likelihoods imply posterior branch $p_0 \in [l, u]$, the robust guide $q_0 = u/(1 + u - l)$ has worst branch ratio $1 + u - l$. Its complete ratio bound must cover unvisited prefixes too.

For universal full support use the *whole-tuple* mixture

$$q(W) = \epsilon\pi_{\text{slot}}(W) + (1 - \epsilon)q_g(W), \qquad 0 < \epsilon \le 1.$$

Draw its component, generate a tuple, and evaluate the actual mixture density, including both terms. Compute

$$Y(W) = \omega_{\text{slot}}(W)/q(W) \le 1/\epsilon, \qquad X_H(W) = Y(W)H(W).$$

Independent tuples satisfy Section 3.2. Accumulate sparse weighted count tails, allocate copies and stop at a declared sample round/certificate or budget. Guide bias has been corrected. The self-normalized estimates are not exactly unbiased at finite sample count.

Variations. Prior-only sampling is $\epsilon = 1$. An evaluable product, quantized-state or positive-contraction law can replace the bit guide. A positive surrogate with log-weight range error Ω supplies concentration bound $\sup P/q \le e^{\Omega}$ without a prior mixture when it already has full support. An exact conditional block may be integrated before sampling the rest; Report 9 H proves the centered unnormalized query variance cannot increase. This may require dense marked queries, so retain an explicit discovery/readout method when conditional integration removes the sampled words.

Cost and limits. Let C_q include drawing and evaluating the full mixture, including any guide construction. Direct cost for S tuples is

$$O(S[C_q + KC_{\text{enc}} + nK] + SK \log(2 + SK)),$$

with $O(SK)$ sparse tail storage. No explicit M-array is required by this execution path, but C_q may include pattern sums and the required sample count can be enormous. A good empirical effective sample size is a diagnostic, not a global unseen-word certificate. Distinct sampling uses the correct legal prior/proposal and its actual exclusions.

10 DEC-06: corrected block moves with global refreshes

In plain terms. Maintain one complete explanation, revise small groups while retaining all outside users in the residual, and occasionally propose a completely new explanation.

Use DEC-05's full-support proposal in an independence Metropolis step:

$$\alpha(W, W') = \min\left\{1, \frac{\omega_{\text{slot}}(W')q(W)}{\omega_{\text{slot}}(W)q(W')}\right\}.$$

With fixed probability $\beta > 0$ choose this global kernel. Otherwise update a fixed-size slot block $\mathcal{B}$ conditional on the current outside messages using $r_{\mathcal{B}} = y - \sum_{i \notin \mathcal{B}} \phi(w_i)$. Exact one-user polar sums or exact K2/K3 joint contractions supply conditional draws. Alternatively a guided block proposal uses its exact conditional Metropolis ratio. Outside messages are current state variables, never supplied truth.

For K2, draw the first word from its exact marginal and the second from its conditional law. For K3, draw one from its exact slot marginal, then a conditional pair. Independently sampling block marginals would lose their dependence. Distinct updates exclude outside words and within-block repeats. Freeze proposal and state-independent block-selection probabilities during a run; otherwise include the correct selection ratios or derive a separate adaptation argument.

Every kernel preserves the exact target. For a certified $\bar{\gamma} \geq \sup_W P_{\text{slot}}(W \mid y)/q(W)$, Report 9 L proves a target minorization $\kappa = \beta/\bar{\gamma}$. Record S states separated by

$$t \geq \kappa^{-1} \log(S/\delta_{\text{mix}})$$

transitions, including before the first record. A coupling bounds their distance from S independent posterior draws by δ_{mix}, giving simultaneous tail error

$$e = \sqrt{\frac{\log(2MK/\delta_{\text{stat}})}{2S}}$$

with total failure at most $\delta_{\text{mix}} + \delta_{\text{stat}}$, capped as necessary. Sparse count frequencies and greedy allocation complete the decoder. An evidence lower bound provides $\bar{\gamma} \leq W_{\max}/Z_-$. Even $Z_- = e^{-(\|y\|+K)^2/(2\sigma^2)}$ is valid, but usually much too conservative.

Cost and limits. Charge St transitions, each local contraction and each global proposal, not just recorded states. Exact K3 still pays repeated pair-operator and alphabet costs. Local-only chains with an unspecified burn-in have an exact invariant law but lack this finite guarantee. The proof does not show that block moves outperform independent sampling. Their role is to explore dependence inside a complete joint decoder, not to declare higher-user effects absent.

11 DEC-07: sequential particles with a complete count guarantee

In plain terms. Preserve several partial explanations, repeatedly allocate more particles to promising ones, and correct the accumulated guidance at the end. Resampling can help exploration, but it creates dependence; the certificate below counts independent batches rather than descendants. Sequential importance/resampling is classical machinery [3]; the finite count-query construction and proof here specify its role in this problem.

11.1 Prefix-growth variant

Let $T = KB$, and let ξ_t be the first t tuple bits. Choose fixed positive guide functions $\ell_t(\xi_t)$ before running a batch, with

$$\ell_0 = 1, \qquad \ell_T(W) = g(y - s_W), \qquad \gamma_t(\xi_t) = \pi(\xi_t)\ell_t(\xi_t).$$

For $0 < t < T$, the guides may be clipped Gaussian, tilted or polynomial continuation estimates. They are intermediate computational weights, not additional physical observations. Use an evaluable legal branch proposal $q_t(b \mid \xi)$, and incremental weight

$$G_t(\xi b) = \frac{\pi(b \mid \xi)\ell_t(\xi b)}{q_t(b \mid \xi)\ell_{t-1}(\xi)}. \tag{7}$$

Without resampling the product telescopes to the exact tuple importance weight.

One batch contains N particles:

1. Start with N empty prefixes of equal weight and $\widehat{Z}_0 = 1$.
2. At each step, independently choose each parent from the previous normalized weights, then draw its next bit from q_t. This is multinomial resampling followed by propagation.
3. Compute $G_t^{(i)}$, their average $\overline{G}_t$, normalized weights $w_t^{(i)} = G_t^{(i)}/(N\overline{G}_t)$, and $\widehat{Z}_t = \widehat{Z}_{t-1}\overline{G}_t$.
4. At $t = T$, retain the weighted complete tuples without an additional resampling step. Set $Y = \widehat{Z}_T$ and $X_H = Y \sum_i w_T^{(i)} H(W^{(i)})$.

All weights are positive. Whole tuple histories, or parent pointers sufficient to reconstruct them, must survive resampling so full-message counts remain available.

11.2 Why independent batches give valid finite confidence

Define $\widehat{\gamma}_t(f) = \widehat{Z}_t \sum_i w_t^{(i)} f(\xi_t^{(i)})$. Conditioning on the previous cloud and averaging the multinomial parents and proposed children gives

$$\mathbb{E}[\widehat{\gamma}_t(f) \mid \mathcal{F}_{t-1}] = \widehat{\gamma}_{t-1}\left(\sum_b \pi(b \mid \xi)\frac{\ell_t(\xi b)}{\ell_{t-1}(\xi)} f(\xi b)\right).$$

Induction from the empty prefix proves

$$\mathbb{E}\widehat{\gamma}_t(f) = \sum_{\xi_t} \gamma_t(\xi_t) f(\xi_t).$$

In particular $\mathbb{E}Y = Z$ and $\mathbb{E}X_H = ZT_H$. This proves unbiasedness of the unnormalized quantities, not of each batch's self-normalized tails.

To obtain a deterministic finite bound, ensure $q_t(b \mid \xi) \geq \epsilon_t \pi(b \mid \xi)$ and $a_t \leq \ell_t \leq b_t$, with $a_0 = b_0 = 1$ and $b_T = 1$. Positive clipping supplies the intermediate a_t, b_t without asserting their accuracy. Then

$$0 \leq X_H \leq Y \leq C_{\text{part}} := \prod_{t=1}^{T} \frac{b_t}{\epsilon_t a_{t-1}}. \tag{8}$$

Run S independent batches and pool their *unnormalized* X_H, Y. Section 3.2 applies with $C = C_{\text{part}}$ and $J = MK$. The pooled positive law weights terminal particles by $Y_s w_{T,s}^{(i)}$. It yields sparse count tails and a feasible greedy action. Equal-weight averaging of batch ratios would not have the displayed unbiased numerator identity.

The bound can be exponentially poor in KB; that is explicit, not hidden as a claimed effective sample size. For fixed N, arbitrarily many independent batches still give consistent queries with a finite confidence prescription. With $N = 1$, the construction reduces to independent sequential importance sampling. This proof does not claim a useful rate for one large correlated cloud.

11.3 A second variant: temper complete tuples

Instead of increasing prefix length, use complete tuples and a fixed schedule $0 = \beta_0 < \beta_1 < \cdots < \beta_T = 1$:

$$\gamma_t(W) = \pi_{\mathrm{slot}}(W) g(y - s_W)^{\beta_t}, \qquad G_t(W) = g(y - s_W)^{\beta_t - \beta_{t-1}} \in (0, 1].$$

Initialize particles from the true source. At each stage weight by G_t, record the mean weight in $\widehat{Z}_t$, resample, and optionally apply any kernel invariant for the normalized γ_t. DEC-06 supplies such conditional/global Metropolis kernels with the stage likelihood. The intermediate exponent changes a computational target only; the final physical noise remains σ^2.

For a marked function f, the same induction now uses $\gamma_{t-1}(G_t K_t f) = \gamma_t(K_t f) = \gamma_t(f)$, by invariance of K_t. At the final stage use the post-resampling/mutation equal-weight empirical measure with $\widehat{Z}_T$, or its equivalent pre-resampling weighted measure if no final mutation is made. Thus independent batches again have unbiased evidence/numerator pairs and now $Y \leq 1$. No equilibrium assumption for each finite mutation run is needed for this identity. Poor exploration still appears as high relative error when Z is small. An adaptive temperature schedule selected from the same particles is outside this fixed-schedule proof.

Costs. Prefix growth needs SNT branch/guide operations, propagation and resampling, plus waveform construction and terminal count reconstruction. Parent pointers require $O(NT)$ records per active batch; cached signals can add $O(Nn)$ storage and $O(n)$ copying work per propagation. Pooled sparse tails require at most SNK updates/storage entries before aggregation. Tempering charges all complete-tuple mutations and stage likelihoods instead. No claim equates SN correlated terminal particles with SN independent posterior samples.

12 DEC-08: compatible quantized-state decoding

In plain terms. Many partial explanations can have almost the same effect on every possible future explanation. Merge those effects while retaining legal count choices and their probabilities.

Use Report 7's compatible future projections before quantizing. A fully specified specialization orders the complete alphabet. At cut i, preserve occupancy k, a representative signal and positive source weight. A transition chooses $a_i = c$, multiplies by $1/c!$ for IID counts, adds $c\phi_i$ and applies Report 9 B's positive representative correction κ_i. Zero-count transitions can be identities. The complete-path weight is

$$\widetilde{\omega}(a) \propto \left(\prod_i \frac{\kappa_i}{a_i!} \right) g(y - \overline{s}_M).$$

Every legal count vector has one path. Merge only states with compatible future constraints; no legal path is dropped merely because its representative has drifted outside a cover of exact signals. Use B's enlarged grid cover or certified constructed ranges.

At occupancy K the future is empty and final scalar absorption is exact. The product likelihood-ratio error telescopes, using the observation once. If the sum of absolute local log errors is at most E_* on every path, then $\mathrm{TV}(P, Q) \le \tanh(E_*/2)$. The occupied-transition schedule charges at most K rounding events and bounds their total error by $\theta K(K-1)/2$, with θ fixed by grid spacings and future coordinate widths.

Record every transition with a count marker $z_{i,c}$. A reverse pass gives $Q(a_i = c) = \partial \log \widetilde{Z} / \partial \log z_{i,c}$. Take tails, allocate and report the TV-derived excess-risk bound. Hold grouping/scalars fixed during this differentiation. A scalar derivative for a message's *mean* alone would be insufficient.

Variations. Exact finalized-resource schedules; cut-specific versus occupancy-fixed grids; state-specific envelopes; backward sampling from the graph followed by DEC-05 correction. The unique-path construction makes its approximate weight evaluable on a proposed count vector. Known alias conditionals may be restored separately. Distinct sampling restricts each transition to $c = 0, 1$ and preserves its future exclusions.

Cost. With at most R representatives per cut/occupancy, there are $O(MKR)$ states and $O(MK^2R)$ transitions, each with its construction/projection cost. The reverse pass needs recorded transitions or charged recomputation, and tails cost $O(MK)$. The covering number can be exponential in projected dimension. This complete option remains explicit in M; a prefix-built state graph that avoids all leaves would require its own transition and readout derivation and is not assumed here.

13 DEC-09: positive regional and feature contraction

In plain terms. Keep exact choices inside manageable regions and simplify how regions interact. Sum the resulting positive model while carrying the total number of users and the requested word counts.

Partition the word alphabet into disjoint regions $\mathcal{R}_r$. A local state is a legal count vector $a^{(r)}$, with occupancy k_r and signal $x_r = \Phi_r a^{(r)}$. Use exact local source factors $1/\prod_w a_w!$ and enforce $\sum_r k_r = K$. Expand the physical likelihood into local unaries and cross terms $\exp(-x_r^T x_s/\sigma^2)$ as in Report 9 C. Declare which edges are retained exactly, deleted with a bound, or represented by positive features.

For a forest, cardinality-indexed sum-product messages give its normalizer and marked local-state marginals. With cycles, condition a sufficient feature/state cutset and sum every sector, or use a specified exact elimination schedule on the approximate factors. At a region incident to several edges, retain their *joint* feature indices. Separate low ranks do not justify multiplying independently marginalized neighbour effects.

For each word in a region, sum local states with $a_w = c$ to obtain count probabilities. Alternatively carry the corresponding count marker through contraction. Normalize, take tails and allocate. Compatible deleted-edge/feature log bounds add to a global range; use $\tanh(\Omega/4)$ or the propagated nonuniform envelopes for its error. Positive Taylor feature order tending upward restores the retained interactions; restoring all edges and states gives exact contraction.

Variations. Regions may be full words, groups of patterns/payloads, or exact constrained blocks. The polar bit/pattern factor graph is another representation, provided all CRC/XOR, pattern, source and occupancy constraints remain exact and the needed count counters cross every separator. Positive low-rank factors, cutset enumeration and sampled cutset sectors are different implementations. Sampling sectors needs their *actual* proposal density and the corresponding importance proof. Backward sampling from a fully normalized positive contraction can feed DEC-05 when the sampled configuration weight/density is evaluable. Local state caps need an omitted-mass certificate.

Cost. Include construction of every local state and every feature, all sector sums, joint separator domain products and marked readout. A small local rank is not a small cyclic contraction. This family has a complete finite procedure but no generally small separator guarantee. Unlike an undeclared loopy-message approximation, it defines one positive count law and a stated error.

One explicit sampled-sector variation avoids any unspecified sector distribution. For L discrete cutset sectors, propose each uniformly and contract its nonnegative evidence Z_ℓ and marked numerator $N_{\ell,H}$. The pair $Y = LZ_\ell$, $X_H = LN_{\ell,H}$ is unbiased for the approximate model's full evidence/numerator. A bound $Z_\ell \le Z_{\text{env}}$ gives $Y \le LZ_{\text{env}}$, so Section 3.2 applies. For example, an upper bound on the complete positive weight divided by the normalized source prior also bounds its evidence and every nonnegative sector. Construct that bound from the factor envelopes. This defines finite sampling and readout without claiming a cheap or concentrated choice of sectors; the approximate model's bias remains separate from the sampling error.

14 DEC-10: connected-component decoding

In plain terms. Compute interactions inside small groups, then assemble every way those groups can coexist among the K users. A two-user cutoff still includes many interacting pairs at once.

Fix one target-K unary reference ν and centered pair factor b, possibly using DEC-02's center. For the positive variant choose a certified floor $0 < b_0 \le b(u,v)$ and put $f_+(u,v) = b(u,v)/b_0 - 1 \ge 0$. In the exact auxiliary-graph expansion, restrict connected components to size at most q. For a queried word or cell D, construct marked connected coefficients

$$C_{m,D}(z) = \mathbb{E}_{\nu^{\otimes m}}\left[z^{A_D} \sum_{\text{connected } G \text{ on } m \text{ slots}} \prod_{(i,j)\in G} f_+(w_i, w_j) \right], \quad m \le q.$$

Assemble all disjoint components by

$$B^{(q)}_{0,D}(z) = 1, \qquad B^{(q)}_{k,D}(z) = \sum_{m=1}^{\min(k,q)} \binom{k-1}{m-1} C_{m,D}(z) B^{(q)}_{k-m,D}(z).$$

Normalize coefficients of $B^{(q)}_{K,D}$ to obtain count probabilities. For a standalone decoder query every full word, take tails and allocate. For selective discovery use these cell queries inside DEC-03. Report 9 E bounds the omitted auxiliary-component mass η_q, giving $\mathrm{TV}(P, Q_q) \le \eta_q$. Order $q = K$ is exact; $q = 1$ is the reference product law. Freeze the target-K reference and floor throughout the recurrence rather than changing them with intermediate block size.

Signed variant. Report 7's expansion with $f = b - 1$ can have smaller coefficients but signed truncations. Carry its connected-tree numerator/evidence remainder and use interval decisions. Negative truncated coefficients cannot be normalized as probabilities. This signed formulation can retain the distinct-source hard pair zeros; the positive-floor product assembly cannot. When combining evidence with other providers, restore every reference-dependent scale factor.

Costs. A direct connected-block construction costs at most $O(\sum_{m\le q} m^2 M^m 2^{\binom{m}{2}})$ arithmetic once pair access is available. Shared marked coefficients and polar K2/K3 operators reduce particular work. The positive per-word assembly costs $O(MK^2q^2)$, and its weighted tree certificate additionally costs $O(KC_{\text{pair}} + K^2M)$. A low cutoff need not give a small error or cheap block construction. The graph-component restriction is not a small *word-list* restriction; its sampling density must not be replaced by the weight of one chosen auxiliary graph.

15 DEC-11: Fourier-based count-query decoding

In plain terms. Express a distribution of waveform sums through frequency functions. Multiplication then replaces convolution, but the result is a signed/complex calculation requiring marked numerator bounds and a positive denominator bound.

15.1 Finite-signature and continuous-frequency variations

The finite-signature route uses Report 7's exact integer signature and its sufficiently large Fourier grid. Keep all modes for the exact query, or selected modes with Report 9 G's omitted-mode bounds. Count-j queries involve the excluded remainder power $K-j$, not just the evidence power K. Strong damping of the evidence modes need not justify dropping the same modes in high-count queries. Constructing likelihood coefficients remains part of the cost.

Alternatively draw $\Omega \sim \mathcal{N}(0, \sigma^{-2} I_n)$. With $\chi_S(\Omega) = \mathbb{E}_{w|S} e^{-i\Omega^T \phi_w}$, exact IID identities are

$$Z = \mathbb{E}_\Omega e^{i\Omega^T y} \chi_{\mathcal{W}}(\Omega)^K, \qquad N_{w,j} = \binom{K}{j} M^{-j} (1-1/M)^{K-j} \mathbb{E}_\Omega e^{i\Omega^T (y - j\phi_w)} \chi_{-w}(\Omega)^{K-j}.$$

Take real parts. Character sums may be explicit, or use Q's certified complex polynomial queries projected onto the unit disk. A one-user error ϵ_χ gives power error at most $h\epsilon_\chi$; include the exact prior count coefficient when bounding the marked term.

Each exact frequency integrand for an evidence or count-tail numerator has real value in $[-1, 1]$. For S independent frequencies and J queries, the simultaneous sampling error is at most $\sqrt{2 \log[2(J+1)/\delta]/S}$, before adding inner errors. This is a signed bounded-variable argument, not the positive-replicate assumption $0 \le X \le Y$ in Section 3.2. Use numerator/event-complement intervals and a certified positive evidence lower bound.

Full decoder and cost. Evaluate all word queries and apply interval allocation, or supply all cell-count queries required by DEC-03 with its simultaneous confidence accounting. A frequency table without word discovery is incomplete. Direct all-word work retains M dependence; each frequency's conditional characteristic and inner degree/pattern work is charged. Increasing modes, sample count and inner accuracy approaches exact queries, but cancellation and small evidence can make the needed absolute accuracy prohibitive. Without-replacement draws require the exact distinct marked coefficients or a controlled source-law comparison; the displayed IID powers do not apply unchanged.

16 DEC-12: signed low-rank query contraction

In plain terms. Approximate large intermediate tables by shorter signed factorizations. Compute a bound on what those factors lose through the same constrained calculation.

Choose a finite factorization/elimination schedule with exact source, CRC, pattern and cardinality constraints. Preserve every needed count counter in its separators. Approximate selected nonnegative likelihood factors f_e by signed low-rank $\widetilde{f}_e$. For a cut table G, a certified operator-norm residual ϵ_r gives $|a^T(G-\widetilde{G})b| \le \epsilon_r\|a\|_2\|b\|_2$, using actual incoming vectors or valid norm bounds.

For multiple factors use constructible nonnegative bounds $|f_e - \widetilde{f}_e| \le \epsilon_e$ and $|\widetilde{f}_e| \le A_e$. The positive error envelope from Report 9 T is

$$\mathcal{E}(x) = \sum_j \epsilon_j(x) \prod_{e<j} (A_e(x) + \epsilon_e(x)) \prod_{e>j} A_e(x).$$

Contract the signed model for $\widetilde{Z}, \widetilde{N}_H$. The two envelope contractions are

$$B_Z = \sum_x \pi(x)\mathcal{E}(x), \qquad B_H = \sum_x \pi(x)|H(x)|\mathcal{E}(x).$$

They certify $|Z - \widetilde{Z}| \le B_Z$ and $|N_H - \widetilde{N}_H| \le B_H$. Normalize intervals, query all words or attach DEC-03, and allocate under Section 3.

Variations and limits. Cut SVDs, signed tensor networks and selected separator compression use the same interface only when their residual certificates are supplied. If $\widetilde{f} = \sum_\ell u_\ell v_\ell$, the positive envelope $A = \sum_\ell |u_\ell||v_\ell|$ gives an explicit contraction, but may inflate its rank/size or magnitude. A full SVD is a finite construction, possibly requiring the very table one hoped to avoid. Include construction, main contraction, every marked envelope and joint separator work. Full rank/zero residual recovers exact contraction. Sampled-entry rank estimates, clipped negative outputs and an unmarked residual alone do not close this decoder.

17 DEC-13: positive-field decoding and exact joint correction

In plain terms. Replace the joint interaction by shared Gaussian auxiliary variables. Conditioned on them, the uncorrected retained-field law separates into one-user categorical laws. Either integrate their count distributions or sample complete tuples together with the fields.

Stack the waveforms as x_W and put $s_W = \sum_k \phi(w_k)$, $t = y/\sigma^2$, $C = (KI - \mathbf{1}\mathbf{1}^T) \otimes I_n/\sigma^2$. Choose a retained covariance $BB^T \preceq C$, whitened field $U \in \mathbb{R}^d$, and define

$$D(W) = x_W^T(C - BB^T)x_W, \qquad c_y = e^{-(\|y\|^2+K^2)/(2\sigma^2)}, \qquad f_k(u,w) = e^{(t+(Bu)_k)^T\phi_w}.$$

The full field has $BB^T = C$ and $D = 0$. For a common retained physical projection, R expresses D as omitted pairwise dispersion and bounds it.

17.1 Conditional-count integration

Ignoring D gives tuple weight $\omega_{\text{slot}}(W)e^{-D(W)/2}$. With exact inner sums $\psi_k(u) = M^{-1}\sum_w f_k(u,w)$, its field density is proportional to $\varphi_d(u)h(u)$, $h = \prod_k \psi_k$, and $p_{k,w} = f_k/(M\psi_k)$. Conditional count probabilities for word or cell A are coefficients of

$$\prod_{k=1}^{K}[1 - p_{k,A}(u) + z\, p_{k,A}(u)].$$

Use a bounded importance proposal from R, calculate its field weight, and integrate these bounded conditional tails. Section 3.2 applies to independent field samples. Enumerate full words for readout, or feed cell queries to DEC-03. The bounded grid construction below gives a deterministic alternative to the field samples.

Direction removal costs at most $\tanh((D_{\max} - D_{\min})/8)$ in TV. Restricting $\|U\| \le R$ adds the corresponding posterior mass bound. Uniform coherent polynomial inner weights add the joint log-ratio bound from Report 9 R. Propagate these in their actual conditional order. Independent normalization of unrelated inner approximations does not necessarily define this joint law.

17.2 A deterministic field integral with an explicit remainder

For the uncorrected retained-field target and a word/cell count-tail indicator H, write

$$F_H(u) = \varphi_d(u)M^{-K}\sum_W e^{t^T s_W + (Bu)^T x_W} H(W), \qquad 0 \le H \le 1.$$

It is evaluable through the inner partitions and conditional count polynomial; F_1 is the evidence integrand, without c_y. Let $a_* = K/\sigma$ and restrict to the cube $[-R,R]^d$. Each component has logarithmic gradient $-u + B^T x_W$, with $\|B^T x_W\| \le a_*$. Since all its summands are nonnegative,

$$F_H(u) \le F_{\max} := (2\pi)^{-d/2} e^{K\|t\| + a_*\sqrt{d}R}, \qquad \|\nabla F_H(u)\| \le (\sqrt{d}R + a_*)F_{\max} =: \Lambda.$$

These are uniform over all count marks in $[0,1]$; differentiating an approximate normalized conditional tail is unnecessary. A midpoint grid with cell side $\Delta = 2R/m$ therefore has integral error at most

$$(2R)^d \Lambda \frac{\sqrt{d}\,\Delta}{2}.$$

The proof bounds the integrand's change from each midpoint by its Lipschitz constant times the cell radius, then sums cell volumes. Node-value intervals add their volume-weighted errors. Multiply by c_y for physical unnormalized weights and normalize with a positive denominator bound.

The field posterior is a mixture of unit-covariance Gaussians with centers of norm at most a_*. Leaving the cube implies Euclidean norm above R, so its omitted mass is at most $\Pr(\|Z_d\| > R - a_*)$ for $R > a_*$. This also bounds the change in count law after restriction. Increasing R and refining the grid gives a finite controlled approximation; it costs m^d nodes and all their inner/count queries. This is a complete quadrature variation, not an efficient high-dimensional integration claim. At $d = 0$ there is no integral.

17.3 Keep the auxiliary variable to make exact correction evaluable

There is a useful complete alternative to evaluating the field-integrated proposal probability of a tuple. Define an unnormalized target on the *joint* space:

$$\Omega(u, W) = c_y \varphi_d(u) M^{-K} \exp\left(t^T s_W + (Bu)^T x_W + \frac{D(W)}{2} \right). \tag{9}$$

Integrating u multiplies by $\exp(x_W^T B B^T x_W/2)$. Adding $D/2$ restores $x_W^T C x_W/2 = (K^2 - \|s_W\|^2)/(2\sigma^2)$, so $\int \Omega(u, W)\, du = \omega_{\text{slot}}(W)$ exactly. This algebra is the reason a reduced field can guide an exact-target sampler.

Choose an evaluable field proposal $q_U(u)$ and independent conditional categorical guides $q_k(w \mid u) > 0$. Draw u, then all K full words, and use

$$Y(u, W) = \frac{\Omega(u, W)}{q_U(u) \prod_k q_k(w_k \mid u)}.$$

No marginal $q(W) = \int q_U \prod q_k$ is needed. This is ordinary importance weighting on an explicit extended space, not an assumption that this marginal density is cheap.

In particular use Q's positive inner polynomials $\rho f_k \le \widetilde{f}_k \le f_k$, with $\widetilde{\psi}_k = M^{-1} \sum_w \widetilde{f}_k$ and $q_k = \widetilde{f}_k/(M\widetilde{\psi}_k)$. Draw each word by exact conditional prefix sums of that *same* polynomial. Then the computable joint weight is

$$Y = c_y \frac{\varphi_d(u)}{q_U(u)} e^{D(W)/2} \prod_k \left[\widetilde{\psi}_k(u) \frac{f_k(u, w_k)}{\widetilde{f}_k(u, w_k)} \right]. \tag{10}$$

Both direction removal and inner guide error are corrected. Their price moves into weight concentration and construction cost. The physical likelihood is still represented exactly once.

17.4 A finite weight bound and full-word readout

Let $\overline{\Sigma}$ be a proved global tilted covariance envelope, $A = B^T(I_K \otimes \overline{\Sigma})B \prec I$, and $D_A = \det(I - A)^{-1/2}$. R supplies $q_U = \mathcal{N}(0, (I - A)^{-1})$ and $\varphi_d h/q_U \le D_A h_0$, $h_0 = \psi(t)^K$. For $0 \le D \le D_{\max}$, (10) therefore obeys

$$Y \le c_y D_A h_0 e^{D_{\max}/2} \rho^{-K}. \tag{11}$$

Use a certified upper bound for h_0 when computing the numerical bound. Since the full evidence is at least $c_y h_0$, the exact concentration ratio is at most $D_A e^{D_{\max}/2} \rho^{-K}$. The sufficient curvature regime, determinant, dispersion and inner degree can each make this unattractive. On an unbounded field domain a rule choosing degree from $\|u\|$ must enforce the same ρ for *every* u, not just verify sampled points.

Outside this Gaussian-envelope regime, restrict u to a ball and propose $q_U = \varphi_d \mathbf{1}_{\|u\| \le R}/p_R$, where $p_R = \Pr(\|Z_d\| \le R)$. For slot-sum-zero retained fields, $h(u) \le \exp(K\|t\| + KR/\sigma)$ on the ball. This gives the finite bound

$$Y \le c_y p_R e^{D_{\max}/2} \rho^{-K} \exp(K\|t\| + KR/\sigma).$$

After the D correction, the conditional field given W is still $\mathcal{N}(B^T x_W, I)$, with $\|B^T x_W\| \le K/\sigma$. Thus its exact joint posterior mass outside the ball is at most $\eta_R = \Pr(\|Z_d\| > R - K/\sigma)$ for $R > K/\sigma$. The sampler targets this restricted joint law; add η_R through the restriction rule. Rejection sampling the base Gaussian until it lies in the ball costs an expected $1/p_R$ attempts.

Independent joint draws now give sparse full-word tails via Section 3.2, followed by greedy allocation. This closes discovery without a dense table of conditional count polynomials. The factor $e^{D/2}$ generally recouples words given u, so one cannot simultaneously apply the uncorrected Poisson-binomial readout to those corrected weights. At full rank with exact inner laws, conditional integration instead gives the earlier Rao–Blackwell query variation. At zero retained rank the joint sampler becomes a corrected one-user product proposal.

Costs and source boundary. Charge field matrices, sampling/rejection, K inner partition and prefix-sampling calculations, every pattern sum, exact waveform/dispersion evaluation and sparse readout. Projection may reduce field dimension while worsening correction concentration. The IID conditional product is essential to these inner formulas. For distinct tuples the identity for a fixed W remains true, but one must use the injective prior, legal proposal and its coupled normalization/bounds. Reusing the IID product integral or the displayed concentration proof is not valid.

18 Combine the designs without hiding a missing operation

18.1 Useful compositions with explicit interfaces

1. **Queries plus selective discovery.** DEC-01, 10, 11, 12 and conditional-field DEC-13 can answer DEC-03's cell-count questions. Their all-word version is already a finite decoder; the prefix wrapper is a possible reduction in query count, not a guarantee that their preprocessing becomes sub-exhaustive.
2. **Approximate law plus exact correction.** DEC-02, 08 and fully normalized positive DEC-09 can propose for DEC-05. Backward sampling, density evaluation and normalizers are required operations. DEC-13's extended-space construction is a separate way to avoid one intractable marginal density. A per-word residual estimate is not automatically a drawable joint law.
3. **Exact local algebra inside global exploration.** Polar K2/K3 contractions supply DEC-06 blocks and DEC-07 tempering mutations or exact conditional integration. All outside-user dependence survives in the current residual or exact outer sum. No assertion that interactions beyond three users are negligible is needed.
4. **Reference choice before interaction approximation.** DEC-02's center can initialize DEC-10 or a guide, while exact parity grouping precedes DEC-09/12 features and exact future projection precedes DEC-08 quantization. Recentring leaves the field dispersion invariant, so its benefit does not transfer automatically.
5. **Moment information plus multiple bounded providers.** P's count LP and prefix consistency can intersect valid moments/numerators from different providers using one common exact evidence. Their approximate point estimates need not form one joint law. The low-order obstruction tells when a new query is required.
6. **Lists and search plus omitted-mass control.** Product search or sampling can suggest candidate words; DEC-04 or A's envelope can bound what was omitted. Restricting an inference engine to that list costs posterior support mass. Alternatively keep full support with a defensive proposal and correct it.

18.2 Mixtures, query selection and adaptation

Several full-support proposals q_r can be combined as $q = \sum_r \lambda_r q_r$ with fixed nonnegative weights summing to one. Draw a component but weight using the *mixture* density, which requires every contributing $q_r(W)$. If each $P/q_r \le \gamma_r$, then

$$q(W) \ge P(W) \sum_r \lambda_r/\gamma_r, \qquad \sup_W P(W)/q(W) \le \left(\sum_r \lambda_r/\gamma_r \right)^{-1}.$$

This is a conditional algebraic bound, not a promise that computing all densities is worthwhile. Incompatible latent spaces cannot be mixed by omitting their marginalization or defining no common target.

Valid deterministic intervals can be intersected. Random intervals need simultaneous coverage across queries, provider choices and stopping rounds. Use conditional fresh-batch guarantees after adapting a guide; the report does not certify arbitrary reuse of adaptive samples. For a final action, use the strongest valid total-tail or direct-gain certificate actually obtained. Do not add two estimates of the same bias, or declare a bias removed merely because a correction was proposed.

18.3 Exact ambiguity and incomplete algorithm names

Restoring source-prior conditionals inside known disjoint equal-signal classes cannot increase TV (Report 7). Charge the class construction and full-word marks. Selecting a preferred word inside an ambiguous class does not remove the physical model's nonidentifiability.

Generic loopy BP, EP, AMP/VAMP, SC/SCL and a neural guide are not additional derived full decoders here. They still need count readout, unseen-word handling and error accounting. A message calculation may guide DEC-05/07; a beam with its correct frontier is a DEC-04 variation. Learning is unnecessary here. Historical D2 remains an empirical comparator, with no new count-posterior or AMP guarantee from this report.

19 Coverage of the exact and approximation work

19.1 Every approximation entry has a defined role

Report 9	Resource	Decoder role
A	Omitted-support evidence	DEC-04 and candidate/restricted variants; keeps discovery loss explicit.
B	Compatible quantization	DEC-08, with optional corrected backward samples in DEC-05.
C	Regional counts / sparse cross edges	DEC-09; exact cardinality survives edge simplification.
D, J	Positive features / cyclic cutsets	DEC-09; joint indices and sector costs remain.
E	Positive connected restriction	DEC-10, full-word or prefix-marked readout.
F, N	Residual moments / exponential tilt	DEC-01 queries, DEC-03/04 bounds, DEC-05/07 guides.
G	Marked Fourier calculation	DEC-11; signed normalization rather than a sampled count law.
H	Weighted samples / conditional integration	DEC-05 and the generalized replicate interfaces in DEC-07/13.
I	Prefix gain and discovery	DEC-03; also a readout wrapper for query-based families.
K	Weighted correction / contrasts / aliases	Corrections in DEC-05/13 and tighter query/decision certificates.
L	Guides and corrected local moves	DEC-05/06; guides and stage kernels in DEC-07.
M, P	Multiplicity moments / consistent LPs	DEC-01/03 and any provider yielding the required marked constraints.
O, R	Positive fields / reductions / integration	DEC-13, including the new joint correction route.
Q	Finite polar inner and residual series	DEC-01/02/03/05/07/11/13; degree and pattern work charged.
S	Variational reference / evidence gap	DEC-02; references and proposals in DEC-05/10.
T	Signed low-rank errors	DEC-12 and its optional DEC-03 discovery wrapper.

Restriction, a moment LP and an alias correction are deliberately not separate decoder families. They modify or certify a full execution path. This table accounts for all A–T without enumerating an unbounded Cartesian product of settings.

19.2 The major exact resources also survive the transition

Report 7 resource	Where it is used
Common posterior, factorial source, count tails	The contract and final readout of every family.
Excluded residual recursion	REF, DEC-01, guided conditional queries and marked Fourier counts.
Inside/outside convolution and marked reverse passes	REF, DEC-08/09; no loss of word identity on merging.
Conditional characters, Walsh and overlap parities	Inner queries, DEC-02 discovery, DEC-06 blocks and DEC-09/10/12 contractions.
Compatible projections and reachable-state bounds	DEC-08 construction and its explicit remaining state cost.
Joint separator/rank obstruction	DEC-09/12 cost account; low physical dimension is not assumed to imply low kernel rank.
Support duals, cached bounds and adverse prior mass	DEC-03/04 certificates and guide/query bounds.
Fourier and connected identities	DEC-11 and DEC-10, with their exact limits retained.
Reference changes, reweighting and exact aliases	DEC-02/05/10/13 and common ambiguity/correction rules.

20 What is now ready to review

The result is a list of complete mathematical decoder options, not a commitment to thirteen independent implementations. A small implementation may share one source/encoder interface, count allocator, interval/replicate readout and a few query primitives across many variations. The important distinctions are what is constructed, what is sampled or summed, where approximation enters, how full messages appear, and what certificate is available.

The additional closure work is:

1. Identical-slot product discovery needs only its highest $\min(K, M)$ word probabilities, with a proved exchange argument and finite conditional-character prefix search. A slot-specific product variation instead has explicit coordinate updates and Poisson-binomial readout, without the same discovery shortcut.
2. Field sampling can retain its auxiliary variable and use a directly evaluable joint weight. This corrects both discarded directions and polynomial categorical guides without evaluating an unavailable integrated tuple proposal. Bounded-domain and curvature-regime guarantees remain explicit.
3. Prefix or tempered particle batches have unbiased unnormalized count queries and finite independent-batch confidence. Correlated particles are not treated as independent observations, and useful concentration remains unproved.
4. Uniform cutset-sector sampling and bounded field midpoint quadrature have explicit proposals or integration remainders, so these named variations also have finite complete constructions.

Completion boundary. Every retained option can return legal full-word counts at a finite resource budget and state its residual error bound, even if that bound is one. The options with convergent refinement are identified; standalone Gaussian and variational product approximations have irreducible modeling-error conditions rather than an automatic exact limit. No option is claimed tractable at the intended load solely because its inner equations are finite.

Review before code. Decide which execution families and variations are worth retaining for the first comparison. For each chosen design, fix its query/proposal provider, exact versus approximate corrections, discovery rule and numerical stopping budget. The subsequent implementation stage should measure actual tail/decision error, certificate tightness or failure probability, and total construction, inference and readout cost separately. This report supplies that choice set; it does not select a winner or start simulations. New derivations should address a specific remaining computational obstacle exposed by a chosen design.

References

[1] Project Report 7, *Polar decoding from the posterior*. Exact report. Exact identities, finite decoders, reductions, obstructions and decision certificates for the fixed contract.

[2] Project Report 9, *Polar Bayesian decoding: approximation algebra*, 19 October 2026. Approximation report. Entries A–T and their completion audit supply the approximation formulas and bounds used here.

[3] P. Del Moral, A. Doucet and A. Jasra, *Sequential Monte Carlo samplers*, Journal of the Royal Statistical Society, Series B 68(3), 411–436, 2006. Paper. Cited for the classical sequential importance/resampling context. DEC-07 gives its own finite prefix/tempering and marked-query proofs; no practical concentration or dimension-scaling result is imported.